6G Weekly Survey — Oct 26 - Nov 01, 2026

15 items across 6 topics, ranked by importance.

6G Physical Layer

Vision--RF Cooperative ISAC for Dynamic V2I Networks: Adaptive Beam Management and Semantic Clutter Suppression

Wenqi~Fan, Ning~Wei, Xiangyu Dong, Ahmad~Bazzi, et al. · arXiv eess.SP · 2026-10-19

★★★★■ Strong ISAC contribution demonstrating practical sensor fusion for V2I networks with bounded overhead and graceful degradation, directly relevant to 6G NTN/vehicular scenarios requiring multi-modal sensing.

This paper develops a vision-RF cooperative ISAC framework for vehicle-to-infrastructure networks that uses camera-based visual priors to guide adaptive beam scanning and clutter suppression. The transmitter allocates bounded scan resources based on visual target probabilities and RF history, switching between communication-for-sensing and sensing-for-sensing modes according to posterior angular uncertainty. At the receiver, visual geometry identifies clutter support for motion-aware filtering that preserves slow-moving targets better than temporal cancellation.

Contribution: The novel integration of visual priors with RF ISAC for joint transmit-side adaptive beam management and receiver-side semantic clutter suppression, with reliability-adaptive resource allocation that degrades gracefully under visual blockage.

Relevance to your work: Directly addresses ISAC beam management with multi-modal sensing and semantic processing—combining your interests in ISAC, beam management, and task-oriented communication with practical resource allocation under sensing/communication tradeoffs.

NTN / satellite

Multi-Timescale Context-Aware Radio Resource Optimization for Harmonious NTN-TN Coexistence: A Digital Twin and Open RAN Enabled Framework

Ecem Izde Kaya, Md. Zoheb Hassan, Georges Kaddoum · arXiv eess.SP · 2026-10-23

★★★★■ Addresses a critical 6G coexistence problem with a practical O-RAN implementation path and demonstrates substantial complexity reduction, though limited to simulation without LEO constellation dynamics or hardware validation.

This paper addresses interference from terrestrial downlink transmissions into LEO satellite receivers by proposing DT-STABLE, an O-RAN framework that uses a digital twin to estimate aggregate interference at satellites without real-time measurements. The authors formulate joint beam activation and user association as an NP-hard optimization problem under RF-chain and interference constraints, then develop an efficient multi-timescale scheduling algorithm. Site-specific 28 GHz simulations demonstrate 89.63% interference satisfaction with 210.96 bps/Hz mean sum rate while reducing beam measurement overhead by 99.71% and achieving 33× speedup over exact solvers.

Contribution: First O-RAN-integrated digital twin framework for proactive NTN-TN interference management that decouples short-timescale resource allocation from mid/long-timescale DT updates, achieving near-real-time performance without exhaustive satellite interference measurements.

Relevance to your work: Directly relevant to your NTN research and interest in AI-native air interfaces—the digital twin approach for interference prediction could inform learned PHY models for satellite-terrestrial coordination and beam management in hybrid networks.

AI-native air interface

Telling Neural Network What We Know: Weight Configuration for RNN and Its Application to NextG Real-Time Neural Receiver

Ummay Sumaya Khan, Lizhong Zheng, Lingjia Liu · arXiv eess.SP · 2026-10-22

★★★★■ Addresses critical practical barriers (latency, pilot overhead, online adaptation) for neural receivers in cellular systems with a theoretically grounded approach and 3GPP validation, though real-world deployment viability remains to be demonstrated.

This paper proposes skip-RNN, a lightweight recurrent neural receiver for MIMO-OFDM that performs online time-domain processing with fixed recurrent/skip branches and adaptive linear readout, avoiding costly backpropagation. The authors configure the network weights using domain knowledge by mapping the RNN structure to stable/causal inverse filters for mixed-phase channels, extracting IIR dynamics from channel statistics for the recurrent branch and handling non-minimum-phase components via the skip branch. Simulations on 3GPP channels show effective performance under extremely low pilot overhead compared to classical receivers requiring extensive channel estimation.

Contribution: A principled weight-configuration framework that exploits the structural correspondence between skip-augmented RNNs and target deconvolution filters, enabling domain-knowledge-informed initialization that eliminates offline training while achieving low-latency online adaptation with minimal pilots.

Relevance to your work: Directly advances AI-native air interface design by bridging classical signal processing insights with neural architectures for practical real-time PHY implementation, addressing key challenges in learned receivers that you work on.

On the Adversarial Vulnerability of Neural Receivers with Unrolled LDPC Decoders

Aakash Saini, Jafar Aco Mohammadi, Wolfgang Gerstacker · arXiv cs.IT · 2026-10-21

★★★★■ Addresses a critical gap in AI-native PHY security by demonstrating that standard adversarial robustness metrics are misleading for unrolled iterative receivers, providing essential guidance for 6G neural receiver design and standardization security considerations.

Neural receivers incorporating unrolled LDPC decoders exhibit robustness to standard adversarial attacks due to gradient masking from tanh saturation in belief-propagation check-node updates, with gradient norm decaying by >4 orders of magnitude across iterations. The authors propose two alternative loss functions—L_BP (per-iteration supervision through early BP iterations) and L_LLR (margin loss on pre-decoder LLRs that bypasses the decoder)—that successfully craft adversarial perturbations increasing BLER by 1–4 orders of magnitude. Results are validated on Sionna NeuralReceiver and DeepRx, with L_LLR showing superior generalization to unseen channels at low perturbation power.

Contribution: First rigorous analysis of adversarial vulnerability in neural receivers with unrolled iterative decoders, identifying gradient masking as the source of apparent robustness and proposing two loss functions that restore attack efficacy while revealing true security gaps.

Relevance to your work: Directly relevant to AI-native air interface development: exposes fundamental security vulnerabilities in learned receiver architectures that must be addressed before deployment, and the gradient-flow analysis informs robust neural receiver design principles for beyond-5G systems.

DIGOM: Disentangled Interaction Graphs for Outcome prediction in Multimodal healthcare

Hakima Laribi, Olivier Lefebvre, Dan Poenaru, Martin Vallières · arXiv cs.LG · 2026-10-23

★■■■■ This is a healthcare machine learning paper with no connection to wireless communications, physical layer design, channel modeling, or any aspect of Beyond-5G/6G research.

This paper proposes DIGOM, a graph-based framework for multimodal healthcare outcome prediction that jointly models three interaction types: intra-patient cross-modal, cross-patient same-modal, and cross-patient cross-modal relationships. The method learns sparse, differentiable multiplex graphs with patient-adaptive connectivity through a learnable sampling mechanism. Evaluated on MIMIC-IV and NIH-TB datasets across five clinical tasks, DIGOM achieves highest AUROC while maintaining sparsity and demonstrating robustness to missing modalities.

Contribution: The novel contribution is the joint disentangled modeling of three complementary interaction types in a single sparse multiplex graph framework with patient-adaptive, differentiable connectivity that enables end-to-end learning without fixed neighbor constraints.

Relevance to your work: tangential

ISAC (sensing + communication)

Array Rotation in Near-Field ISAC: Communication, Detection, and Estimation Under Low-Complexity Transmit Architectures

Marziyeh Soltani, Mark F. Flanagan · arXiv cs.IT · 2026-10-22

★★★★■ Addresses emerging near-field ISAC challenges for 6G with practical architectures and exact propagation modeling, providing actionable design insights for rotatable arrays which may become relevant as near-field operation becomes more common at mmWave/sub-THz.

This paper analyzes near-field ISAC performance when the base-station array can physically rotate, deriving Cramér-Rao bounds and optimal rotation angles for two low-complexity transmit architectures (single-stream superposition and dual-waveform split). The study characterizes how rotation affects communication rate, target detection probability, and parameter estimation accuracy under spherical-wave propagation, showing that single-stream designs can align communication and detection objectives while dual-waveform architectures face an inherent tradeoff. Numerical results demonstrate significant gains over fixed orientation, particularly for range estimation, and validate second-order approximations for optimal angles.

Contribution: First rigorous analysis of array rotation as a degree of freedom in near-field ISAC, providing closed-form CRB decompositions and optimal rotation strategies for practical low-complexity transmit architectures under exact spherical-wave models.

Relevance to your work: Directly relevant to your ISAC research interests; the low-complexity transmit architecture analysis and communication-sensing tradeoff characterization connects to practical 6G system design considerations, though the physical rotation mechanism may be less central than algorithmic beamforming approaches.

Low-Complexity Wideband Channel Estimation for ISAC in Analog Millimeter-Wave Systems

Hamza Touati, Raffaele Viterbo, Mattia Brambilla, Sandra Roger, et al. · arXiv eess.SP · 2026-10-23

★★★■■ Solid incremental work addressing a real computational bottleneck in analog mmWave ISAC beam training, with demonstrated localization performance but limited to narrowband hybrid architectures and ray-traced validation.

This paper extends the Transformed Spatial Domain Channel Estimation (TSDCE) method to wideband mmWave ISAC systems by decoupling angle and delay estimation to reduce complexity. Two variants are proposed: TSDCE-TD (Dict) uses a 1D delay dictionary while TSDCE-TD (FFT) applies zero-padded FFT interpolation, both refined iteratively via successive interference cancellation. Simulations show the method outperforms OMP baselines under practical dictionary sizes and achieves sub-20 cm indoor localization accuracy with substantially lower computational cost.

Contribution: The novelty lies in decoupling the traditionally joint angle-delay sparse recovery problem into sequential stages with FFT-based interpolation, achieving practical complexity reduction while maintaining ISAC localization accuracy below 20 cm.

Relevance to your work: Directly relevant to your ISAC and mmWave beam management interests, particularly for resource-constrained analog beamforming systems where dictionary-based channel estimation complexity is a practical deployment barrier.

Cross-Region Cooperation of UAV Swarms Towards Air-Ground ISAC via Hierarchical Multi-Agent Reinforcement Learning

Linghui Miao, Nhan Thanh Nguyen, Shijian Gao · arXiv eess.SY · 2026-10-22

★★★■■ Solid incremental work on UAV-enabled ISAC with practical hierarchical control structure, though application scope is narrow and multi-UAV ISAC benefits are well-known.

This paper addresses UAV swarm coordination for air-ground ISAC in multi-region scenarios where vehicle demand is unevenly distributed. A hierarchical multi-agent reinforcement learning framework coordinates slow-timescale cross-region UAV reallocation to match communication resources with demand, and fast-timescale intra-region control of UAV positions and transmit power. For sensing, it combines observations from multiple regions and uses marginal Cramér-Rao bound to guide adaptive phase synchronization handshaking between UAVs.

Contribution: The hierarchical MARL framework decouples inter-region resource balancing from intra-region UAV optimization while introducing adaptive handshaking for multi-UAV cooperative sensing guided by marginal CRB utilities.

Relevance to your work: Relevant as it combines ISAC with hierarchical MARL for resource allocation and introduces CRB-based optimization for cooperative sensing, which connects to your interest in ISAC system design and AI/ML approaches for next-generation wireless.

6G Demos & Trials

From Data to Deployment: A Scoping Review of Empirical Tools, Techniques, and Operational Readiness in Data-Driven 5G/6G Network Optimization

Muhammad Kabeer, Rosdiadee Nordin, Sian Lun Lau · arXiv eess.SP · 2026-10-19

★★★■■ Valuable meta-analysis addressing reproducibility and deployment gaps in AI/ML for wireless, but methodological contribution rather than technical innovation in AI-native air interface design.

This scoping review analyzes 72 journal studies (2022-2026) on AI/ML for 5G/6G network optimization, examining empirical evidence types, evaluation methods, and operational readiness. It finds that 51% use real-world/testbed data, 35% combine measured and simulated data, and 14% use only synthetic data, with most (63%) conducting offline analysis. The authors propose a six-module EFOR checklist to standardize evaluation and reporting practices across the research community.

Contribution: Systematic classification of empirical evidence quality and evaluation methodologies in AI-driven network optimization research, introducing a structured reporting framework (EFOR) to bridge the gap between research demonstrations and operational deployment.

Relevance to your work: Directly relevant for ensuring your AI-PHY and learned receiver work translates to deployment, particularly the emphasis on validation methodology and distinguishing synthetic vs. real-world evidence in channel modeling and neural transceiver evaluation.

K-AoI: A Spatio-Temporal Synchronization Metric for Digital Twin-based Teleoperation in Dynamic Network Environments

Lu Tian, Zhenhui Yuan, Zexin Deng, Wenjuan Zhang · arXiv cs.NI · 2026-10-19

★★★■■ Solid contribution addressing real operational needs in digital twin verticals with rigorous experimental validation and open-source release, though focused on application layer rather than PHY/MAC innovation.

The paper introduces Kinematic Age of Information (K-AoI), a spatio-temporal synchronization metric for digital twin teleoperation that weights standard Age of Information (AoI) by instantaneous Cartesian velocity to capture spatial divergence caused by network delays. Evaluated on a 6-DOF manipulator testbed across seven network conditions (including satellite links), K-AoI shows low rank-correlation with AoI ($\rho \leq 0.65$, dropping to 0.22 under satellite), indicating it captures velocity-dependent effects invisible to time-only metrics. Analysis of AoI tail behavior reveals standard Generalized Pareto Distribution models fail due to deterministic forwarding and bounded jitter in real network conditions.

Contribution: K-AoI is the first metric to translate temporal network staleness into physically meaningful spatial error by incorporating robot kinematics, providing a task-relevant synchronization measure for digital twin systems that time-based AoI cannot capture.

Relevance to your work: Tangential—addresses cross-layer quality metrics for digital twin use cases in 6G verticals, but does not engage with AI-native air interfaces, physical layer design, or wireless system architecture that forms your core research focus.

Experimental Classification of xApp Conflicts in O-RAN: Slicing vs. Per-UE Scheduling

Rashmi Sharan Sinha, Merim Dzaferagic, Marco Ruffini, Nima Afraz · arXiv cs.NI · 2026-10-23

★★★■■ Solid experimental work addressing a practical O-RAN deployment challenge with real hardware validation, though the specific slicing conflict taxonomy is narrow and not directly tied to 6G physical-layer innovation.

This paper experimentally demonstrates control-intent conflicts in O-RAN Near-RT RIC when a slice-level PRB allocation xApp interacts with per-UE scheduling on a 5G testbed (srsRAN gNB, Open5GS core, USRP X310, COTS UEs). A utilization-aware slicing controller that dynamically reallocates PRBs from underutilized to saturated slices logged 147 floor–cap violations, 49 budget over-subscriptions, and 11 oscillation episodes over 276 cycles, yet reduced unmet demand by 12–33% and idle allocation by 21–23% versus static slicing. The work provides the first reproducible experimental characterization of hierarchical resource-control conflicts in real O-RAN hardware.

Contribution: First experimental demonstration on real 5G hardware of cross-granularity control-intent conflicts between slice-level and per-UE resource allocation in O-RAN RIC, with quantified conflict patterns and performance trade-offs.

Relevance to your work: Tangential—focuses on RAN resource-management orchestration conflicts rather than AI-native air interface design, PHY-layer learning, or advanced waveforms for 6G systems.

Controlled Radio Fault Injection and Telemetry Evaluation for Open 5G RANs

Babak Azkaei, Maxim Bîtiu, Georgios Exarchakos, Kishor Chandra Joshi · arXiv cs.NI · 2026-10-22

★★★■■ Solid engineering contribution for testbed methodology; useful for validating RAN monitoring/ML systems but limited to software baseband scenarios and incremental over existing srsRAN fault injection approaches.

The paper presents a controlled fault injection framework for software-based 5G RANs (srsRAN) that applies scheduled impairments (uplink interruptions, additive noise) to I/Q samples via ZeroMQ channel brokers while recording native and programmable telemetry alongside application-layer metrics. Through 45 trials, they demonstrate that native telemetry (e.g., missing feedback events) can retrospectively identify interruptions but noise-induced decoding errors do not trigger such events, with mean 95th-percentile delays increasing 354-604 ms for interruptions and 113 ms for noise. The framework enables systematic evaluation of RAN telemetry and monitoring strategies under known radio impairments before deploying them in operational or ML-based systems.

Contribution: A reproducible testbed workflow that synchronously injects controlled I/Q-level impairments and collects multi-layer telemetry, enabling empirical validation of which RAN measurements can distinguish specific radio fault types from their service impacts.

Relevance to your work: Relevant for validating AI/ML physical-layer solutions: this controlled testbed methodology could be adapted to evaluate learned receivers or neural channel estimators under known impairments, bridging simulated fault models to real protocol dynamics.

SILMARILS-MV: Information-Theoretically Secure Multi-Verifier Signatures with Dealer-Free Key Issuance and Linear Aggregation for Permissioned Blockchains

Khadijeh Bagheri, Hassan Khodaiemehr, Chen Feng · arXiv cs.IT · 2026-10-22

★★■■■ Addresses post-quantum security for permissioned blockchains with impressive performance gains, but limited to permissioned settings and does not intersect with wireless PHY, air interface design, or radio resource management relevant to 6G.

The paper presents SILMARILS-MV, an information-theoretically secure multi-verifier signature scheme for permissioned blockchains that achieves unconditional security without trusted dealers or quantum channels. Keys are issued via dealer-free MPC among validators, signatures are linearly aggregatable vectors, and

threshold notarization produces constant-size finality certificates (32(ω+1) bytes). In a 100-validator HotStuff prototype, the scheme achieves 2.8x higher throughput than post-quantum ML-DSA-44 with 67x less CPU and 22x less authentication traffic at 1000 validators.

Contribution: First information-theoretically secure signature scheme for blockchains that combines dealer-free key issuance, linear aggregation, multi-hop transferability, and constant-size certificates without requiring quantum channels or trusted authorities.

Relevance to your work: Tangential—this is cryptographic protocol work for distributed ledgers; no connection to AI-native air interfaces, channel modeling, CSI processing, ISAC, NTN, semantic communications, or 6G waveforms.

Detection and Validation of Hidden Subgroup Disparities in Clinical AI

Alceu Bissoto, Trung-Dung Hoang, Tim Flühmann, Nina Weng, et al. · arXiv cs.LG · 2026-10-23

★■■■■ This is healthcare ML fairness work with no connection to wireless communications, physical layer design, or 6G system architecture.

This paper evaluates algorithmic subgroup discovery methods for detecting hidden performance disparities in clinical ML models, specifically applied to cardiomegaly classification in chest X-rays. Unlike traditional metadata-based analyses that rely on predefined patient demographics, the approach discovers coherent subgroups from learned feature representations, revealing larger performance gaps (accuracy range 0.61-0.90) than conventional methods (0.80-0.94). The authors demonstrate that discovered subgroups are semantically interpretable and their annotations can be used to mitigate exposed biases.

Contribution: The work provides systematic evaluation of unsupervised subgroup discovery as a practical performance monitoring tool for clinical ML, demonstrating it can uncover hidden disparities that metadata-based analysis misses when ground-truth stratifications are unknown.

Relevance to your work: tangential

Semantic communication

Secure Agentic AI and Multimodal Intelligence in 6G Communication Systems

Satyam Gaba, Devanshu Motwani, Siva Sai, Vineet Suri, et al. · arXiv eess.SP · 2026-10-21

★★★■■ Timely treatment of emerging security concerns as 6G architectures increasingly rely on autonomous AI agents, though primarily a survey organizing existing knowledge rather than proposing novel defenses.

This paper surveys security and privacy challenges introduced by agentic AI systems in 6G networks, where autonomous agents reason over multimodal data, use network tools, and coordinate across distributed entities for applications like semantic communication and digital twins. The authors present a system-level view spanning perception, memory, tool use, and multi-agent orchestration, then taxonomize attacks across sensing, cognition, action, memory, and communication layers. Mitigation strategies are reviewed including prompt defenses, inter-agent trust mechanisms, and privacy-preserving learning frameworks.

Contribution: First comprehensive security-focused survey mapping agentic multimodal AI vulnerabilities to 6G system layers, providing structured taxonomy of attack surfaces beyond conventional network security models.

Relevance to your work: Directly relevant to semantic communication deployment and AI-native air interface security, particularly for understanding vulnerabilities in learned PHY systems and multi-agent coordination in distributed ISAC or NTN scenarios.